

SMART CONTRACT SECURITY AUDIT - VAULTBANK

One-Page Visual Audit Report

Purpose: ETH vault with per-user balances and withdrawal function.

Risk	Severity	SWC	OWASP
withdraw() performs external call before decrementing balance	Critical	SWC-107	SC08 Reentrancy
No reentrancy guard on ETH withdrawals	High	SWC-107	SC08 Reentrancy
No owner/admin recovery or insolvency controls for unexpected ETH flows	Low	N/A	SC02 Financial Risks
deposit() accepts zero-value deposits	Low	N/A	SC05 Input Validation

Root Cause: The contract violates checks-effects-interactions by transferring ETH before updating the user's balance.

Business Logic Findings: A malicious receiver can reenter withdraw() while the balance is still unchanged and drain more ETH than intended if liquidity exists.

BENCHMARK METRICS

Time_seconds: 75 | Confidence: 0.97 | Total_findings: 4

Critical: 1 | High: 1 | Medium: 0 | Low: 2

Precision_estimate: 0.95 | Coverage_estimate: 1.00 | Risk_score: 90/100

GOLD STANDARD EVALUATION

TP: N/A | FP: N/A | FN: N/A | Precision: N/A | Recall: N/A | F1: N/A